Unified Military Analytics and Comparison Dashboard-DV

Project Statement

This project aims to develop a fully interactive dashboard suite for analyzing global military power in 2025. Using Python, military metrics will be scraped from GlobalFirepower.com for 140+ countries via a pre-defined list (links_for_military_data.txt). The project leverages Python-based data pipelines to scrape, clean, and process open-source defense data from GlobalFirepower.com

Unlike traditional single-platform dashboards, GDM emphasizes cross-platform flexibility — allowing deployment in tools like Tableau, Power BI, or Python-based web apps built using Streamlit or Dash.

The platform unifies more than 50+ defense and economic indicators across four interconnected modules:

- **Quick Stats** – A global overview of rankings, trends, and highlights
- **Nation Overview** – Detailed analysis of individual country capabilities
- **Compare Powers** – Side-by-side military and economic comparison
- **Coalition Builder** – Interactive simulation of alliance strength and combined assets

The final deliverable is a **comprehensive analytics suite** that enables users to compare nations, evaluate coalitions, and analyze KPIs like **Power Index Rank Gap**, **Assets per Capita**, and **Defense Budget-to-GDP Ratio** through intuitive and dynamic visualizations.

Outcomes

- Fully functional scraping system using pre-provided URLs from the link : https://drive.google.com/file/d/1G6aBlH5-bx1x4SsrcNsw-YoxEuQccLwK/view?usp=sharing
- Cleaned and standardized dataset covering **140+ countries and 50+ indicators**
- KPIs derived using custom Python scripts
- Four cross-linked dashboards (Tableau / Power BI / Streamlit / Dash or any frontend)
- Unified visualization ecosystem with region filters, dropdowns, and navigation controls
- A GitHub-hosted deliverable containing scripts, data, and documentation

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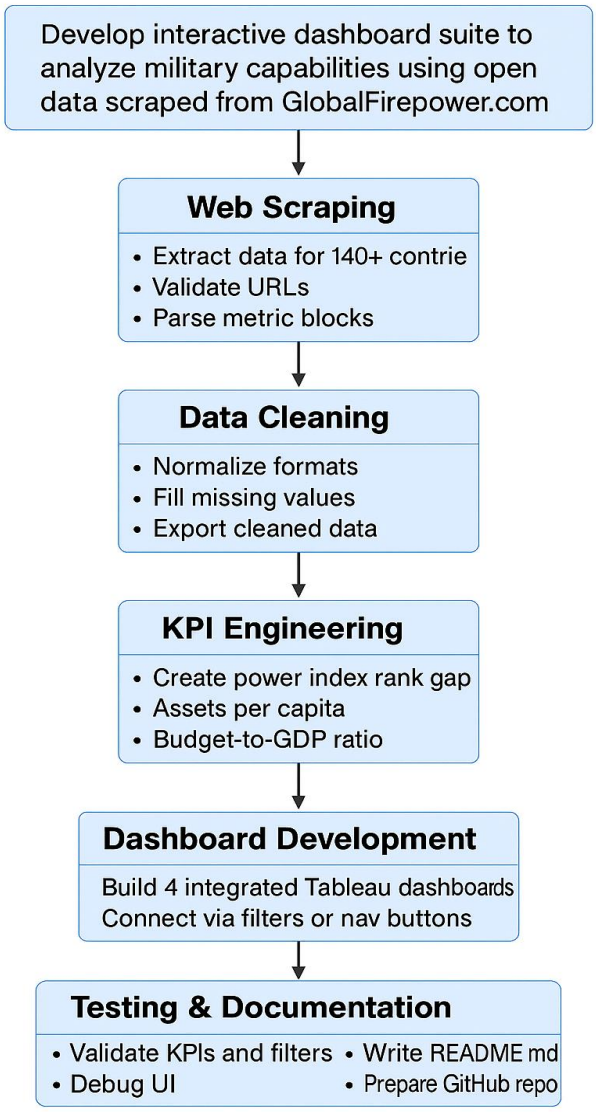
Modules to be implemented

- Web scraping using requests, BeautifulSoup
- Data cleaning and structuring
- KPI feature engineering
- Dashboard development and parameter linking
- Final integration into a seamless user experience
- Documentation and GitHub packaging



Milestone 1: Data Collection and Preparation (Weeks 1–2)

Module 1: Scraping Setup and Execution

Tasks:

- Use links_for_military_data.txt from the drive link as source of all URLs
- Scrape country-level military metrics (e.g., total aircraft, tanks, budget)
- Store raw data into structured CSV files
- Save per-country HTML if needed for debugging

Deliverables:

- Script: scrape_military_metrics.py
- Output: military_raw_data.csv

Evaluation:

- ≥ 95% URL success from the provided list
- Metric blocks correctly parsed and saved for 140+ countries

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Module 2: Data Cleaning and Structuring

Tasks:

- Clean text: remove commas, %, +, special symbols
- Convert metrics to numeric formats
- Standardize column names (e.g., total_aircraft, active_personnel)
- Handle missing/null values
- Save a clean dataset for Tableau input

Deliverables:

- military_cleaned.csv
- Python notebook: clean_data.ipynb

Evaluation:

- < 2% missing/null after cleaning
- No structural errors in the output dataset

Milestone 2: KPI Engineering and Tableau Prep (Weeks 3–4)

Module 3: KPI Feature Engineering

Tasks:

- Compute:
    - Power Index Rank Gap

    - Assets per Capita
    - Budget-to-GDP Ratio
- Enrich with metadata: region, continent, alliance flags (e.g., NATO)
- Format in wide and long forms for Tableau use

Deliverables:

- military_final.xlsx
- Script: generate_kpis.py

Evaluation:

- All KPIs present and correctly calculated
- Data loads in Tableau without transformation

Module 4: Dashboard Planning and Prototyping

Tasks:

- Draft layout sketches or wireframes for:
    - Quick Stats
    - Nation Overview
    - Compare Powers
    - Coalition Builder
- Define filters, buttons, and navigation logic
- Build a prototype dashboard using sample data

Deliverables:

- Storyboard for dashboard layouts
- A dashboard application prototype.

Evaluation:

- Interactions and KPIs are mapped correctly
- One dashboard (Quick Stats or Nation Overview) operational

Milestone 3: Full Dashboard Development (Weeks 5–6)

Module 5: Build Quick Stats and Nation Overview Dashboards

Tasks:

- Quick Stats:
    - Show top 10 countries by Power Index
    - Add filters: Region, Continent, Alliance
    - KPI cards with dynamic values
- Nation Overview:
    - Select country to display full profile
    - Use bar/radar charts for metrics
    - Add tooltips with rank or comparisons

Deliverables:

- Sample application of dashboard
- Fully built Quick Stats and Nation Overview tabs

Evaluation:

- Country/region filters work as expected
- Nation Overview visualizes all major metrics

Module 6: Build Compare Powers and Coalition Builder Dashboards

Tasks:

- Compare Powers:
    - Side-by-side comparison of any 2 countries
    - Metrics: manpower, aircraft, navy, budget, KPIs
    - Use Tableau parameters for country selection
- Coalition Builder:
    - Multi-country selector (via filter or highlight)
    - Show aggregated totals for coalition metrics
    - Compare selected coalition against another or a reference country
- Integrate Dashboards:
    - Add navigation buttons (e.g., from Quick Stats to Compare Powers)
    - Link all filters across dashboards

Deliverables:

- Final Tableau workbook: global_military_firepower_2025.twbx
- All 4 dashboards fully integrated

Evaluation:

- Seamless transitions between dashboards
- Compare and Coalition dashboards work with all inputs

Milestone 4: Final Review and Delivery (Weeks 7–8)

Module 7: Testing and Debugging

Tasks:

- Test all filters, parameters, and navigation across dashboards
- Verify accuracy of all values displayed (manual spot checks)
- Fix layout bugs, tooltip issues, or mislabelling
- Ensure workbook runs smoothly in Tableau Public/Desktop

Deliverables:

- Debugged and polished workbook
- QA checklist

Evaluation:

- No functional bugs
- Navigation and data display verified end-to-end

Module 8: Documentation and GitHub Release

Tasks:

- Write final README covering:
    - Scraping method
    - KPI definitions
    - How to open/use the Tableau dashboards
- Organize project files: /scripts, /data, /dashboard, /docs
- Push final version to GitHub
- Optional: Publish dashboard to Tableau Public

Deliverables:

- GitHub repository with all files
- Final README + dashboard usage guide

Evaluation:

- Repo is clean, documented, and shareable
- Dashboard is ready for public or portfolio use

Evaluation Criteria

Milestone	Focus Area	Metric	Target
1	Scraping & Cleaning	Raw & clean data coverage	≥ 140 countries, < 2% missing data
2	KPI Engineering	KPI accuracy and format	5+ KPIs correctly computed
3	Dashboard Development	4 dashboards built & connected	Full integration, navigation OK
4	Delivery & QA	Final packaging & usability	No bugs, GitHub ready

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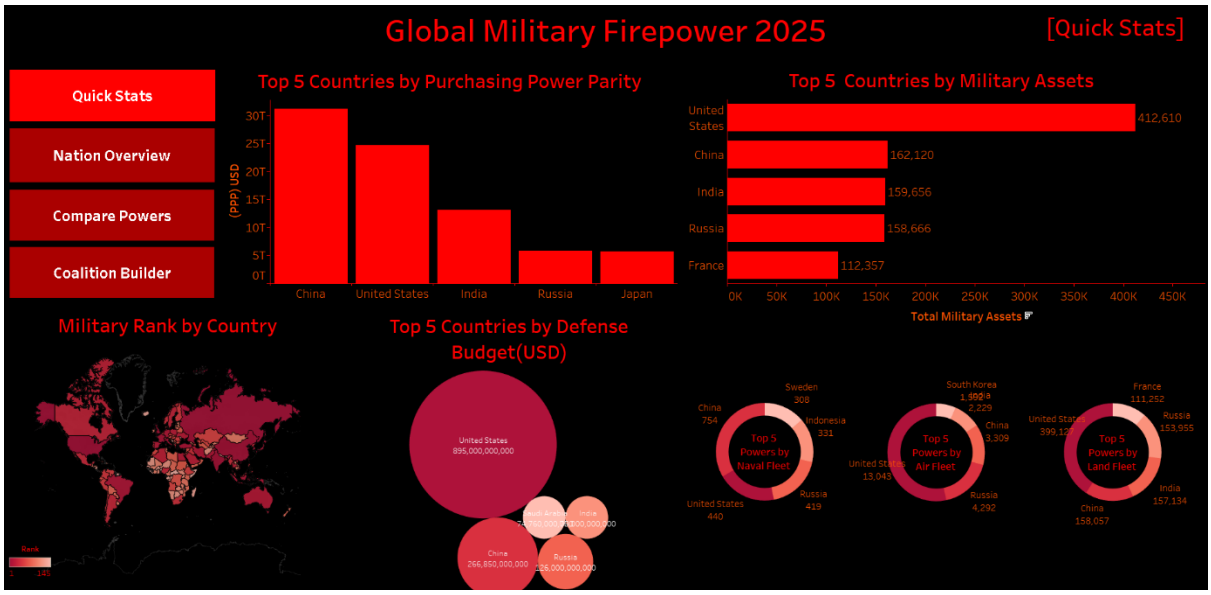
Tech Stack

Area	Tools / Libraries
Scraping	Python, requests, BeautifulSoup, links_for_military_data.txt
Processing	pandas, numpy
Visualization	Tableau Public/Desktop,streamlit,bowerbi,dash
Integration	parameters, filters, buttons
Documentation	Markdown, GitHub

Sample Dashboard Idea :